

This document has some practice questions to help you prepare for the midterm.

1 Derivatives

1.1 Differentiation

1. Use the limit definition to compute the derivative of $f(x) = \frac{x+1}{2-x}$.
2. Use the limit definition to compute the derivative of $f(x) = \cos(3x)$.
Try to simplify your answer as much as possible, you may need this trigonometric identity $\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$.
3. Use the limit definition to compute the derivative of $f(x) = \frac{a}{x}$.
4. Which of the following functions does not have a **continuous** derivative?
a) $x^2 - 2x + 1$, b) $\cos(x^2)$, c) $|x - 1|$

1.2 Total derivative

1. Compute the total derivative of the Beattie-Bridgeman equation of state for the pressure,

$$p(T, \nu) = \frac{RT}{\nu^2} \left(1 - \frac{c}{\nu T^3} \right) (\nu + B) - \frac{A}{\nu^2}, \quad (1)$$

where A and B are some functions of the molar volume ν , here we will consider them as constants.

2. What is the total derivative of a linear model of the form,

$$f(\mathbf{x}) = \sum_{i=1}^m w_i x_i, \quad \text{where} \quad \mathbf{x} = [x_1, x_2, \dots, x_m]. \quad (2)$$

3. What is the total derivative of a polynomial expansion of the form,

$$f(x) = \sum_{i=1}^m w_i x^i \quad \text{where} \quad x^i \quad \text{is } x \text{ to the power of } i. \quad (3)$$

1.3 Partial derivatives

1. The virial expansion is a model of thermodynamic equations of state, and it is given by,

$$P(V_m, T) = \frac{R}{T} \left(1 + \frac{B(T)}{V_m} + \frac{B(T)}{V_m^2} + \cdots \right), \quad (4)$$

where V_m is the molar volume.

Compute the partial derivate of $P(V_m, T)$ with respect to the molar volume.

2. Compute the partial derivative of $f(x, y) = (2x + 6y)^4$ with respect to the variable y

$$\frac{\partial f(x, y)}{\partial y} = \frac{\partial}{\partial y} (2x + 6y)^4 \quad (5)$$

3. Compute the partial derivative of $f(x_1, x_2) = \frac{e^{x_1}}{e^{x_1} + e^{x_2}}$ with respect to the variable x_1

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = \frac{\partial}{\partial x_1} \left(\frac{e^{x_1}}{e^{x_1} + e^{x_2}} \right) \quad (6)$$

2 Integrals

2.1 Substitution

1. Compute the indefinite integral,

$$\int x^3 (1 - 12x^4)^{1/8} dx. \quad (7)$$

2. Compute the indefinite integral,

$$\int \frac{x}{\sqrt{8 - 2x^2}} dx. \quad (8)$$

3. Compute the indefinite integral,

$$\int 7x^4 e^{x^5} dx. \quad (9)$$

4. Compute the indefinite integral,

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx. \quad (10)$$

2.2 Rational functions and logarithms

1. Compute the indefinite integral,

$$\int \frac{x+5}{x} dx. \quad (11)$$

2. Compute the indefinite integral,

$$\int \frac{x}{x+5} dx. \quad (12)$$

Try to rewrite the integrand before integrating, you may need to write $\frac{x}{x+5} = 1 - \frac{5}{x+5}$.

3. Compute the indefinite integral,

$$\int \frac{1}{x \ln(x)} dx. \quad (13)$$

2.3 Trigonometric integrals

1. Compute the indefinite integral,

$$\int \cos^2(x) \sin(x) dx. \quad (14)$$

2. Compute the indefinite integral,

$$\int \frac{\cos(x)}{\sin^3(x)} dx. \quad (15)$$

3. Compute the indefinite integral,

$$\int \sec^9(x) \tan(x) dx. \quad (16)$$

Try to simplify the integrand first, you may need the derivative $\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$.